1. (25 points) Let $x = (x_0, \cdots, x_n), y = (y_1, \cdots, y_n) \in \mathbb{R}^n$. Define

$$d_1(x,y) = \left(\sum_{i=1}^{n} |x_i - y_i|^2 \right)^{\frac{1}{2}}, \quad d_2(x,y) = \left(\sum_{i=1}^{n} |x_i - y_i|^3 \right)^{\frac{1}{3}}$$

   Then d_1, d_2 are two metrics on $\mathbb{R}^n$. Let τ_1 be the topology on $\mathbb{R}^n$ induced by d_1 and let τ_2 be the topology on $\mathbb{R}^n$ induced by d_2. Prove that $(\mathbb{R}^n, \tau_1)$ is homeomorphic to $(\mathbb{R}^n, \tau_2)$.

2. (25 points) Let $Y = \{z \in \mathbb{C} || z| \leq 1\}$ be the unit disk and let $A = [z \in \mathbb{C} \mid |z| = 1\} \subset Y$ be the boundary circle. Let $X = \{z \in \mathbb{C} \mid |z| = 1\}$ be the unit circle and define a map $f : A \to X$ by $f(z) = z^2$, Let $Z = X \cup_f Y$

   (1) (23 points) Prove that Z is compact and connected.

   (2) (2 points) In fact, we have studied the space Z in the class. What is the name of the space Z? (You do not need to give a proof.)

3. (25 points) Let $X = \{(x,y,z) \in \mathbb{R}^3 \mid x^2 + y^2 + z^2 = 1\}$ be the unit sphere and let $S^1 = \{e^{i\theta} \mid 0 \leq \theta < 2\pi\}$. Then S^1 is a topological group under the multiplication $e^{i\theta_1} \cdot e^{i\theta_2} = e^{i(\theta_1 + \theta_2)}$. Define an action of S^1 on X by

$$e^{i\theta}(x,y,z) = (x\cos\theta - y\sin\theta, x\sin\theta + y\cos\theta, z)$$

   Prove that the quotient X/S^1 is homeomorphic to the interval $[-1,1]$.

4. (20 points) Let $\mathbb{C}^* = \mathbb{C} \setminus \{0\}$.

   (1) (7 points) Compute the fundamental group $\pi_1(\mathbb{C}^*)$.

   (2) (7 points) Define a map $f : \mathbb{C}^* \to \mathbb{C}^*$ by $f(z) = z^n$ where 0 is an integer. Compute the homomorphism $f_* : \pi_1(\mathbb{C}^*) \to \pi_1(\mathbb{C}^*)$.

   (3) (6 points) Define a map $g : \mathbb{C} \to \mathbb{C}^*$ by $g(z) = e^z$. Let $\gamma : [0,1] \to \mathbb{C}^*$ be a path with $\gamma(0) = 1$. Prove that there exists a unique path $\tilde{\gamma} : [0,1] \to \mathbb{C}$ such that $\tilde{\gamma}(0) = 0$ and $g \circ \tilde{\gamma} = \gamma$.

5. (5 points) Let $X = \{(x,y) \in \mathbb{R}^2 \mid (x-1)^2 + y^2 \leq 1\} \cup \{(x,y) \in \mathbb{R}^2 \mid (x+1)^2 + y^2 \leq 1\}$. Equip X the subspace topology from $\mathbb{R}^2$. Let $f : X \to X$ be a continuous map. Prove that there exists $x \in X$ with $f(x) = x$.